

Aditya Balakrishnan

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Software Engineer specializing in agentic AI and scalable backend systems — I build AI-agent infrastructure end to end, bring an eval-first discipline to shipping LLM features in production, and default to AI-in-the-loop on every task.

WORK EXPERIENCE

BlinqIO

Oct. 2024 – Present

Software Engineer, R&D

Bengaluru, India

- Engineered the core AI engine – a closed-loop, multi-agent system (planner-executor, built on Mastra) that orchestrates foundational models (Claude, OpenAI) to turn natural-language intent into reliable browser automation, with observation-grounded verification that grounds assertions in real page state to eliminate hallucinated checks.
- Built the platform's LLM evaluation suites that gate every model and prompt change in CI – multimodal vision-language-model (VLM) judges plus property-based tests – quantifying accuracy, latency, and cost trade-offs before anything ships to production.
- Established the team's agentic-engineering playbook – custom Claude Code plugins/skills, **AGENTS.md** standards, and shared agent workflows – and use LLM agents daily to accelerate bug triage and session analysis.
- Architected security-hardened, sandboxed execution of untrusted user code with OS-level process isolation and capability-scoped APIs (strict trust boundaries, TLS-required connections, host allowlists), eliminating a custom-code blocker and enabling a customer migration to complete at 100% parity.
- Built type-safe backend microservices (Node.js, Express, tRPC) over PostgreSQL and MongoDB, and founded the 20+ package monorepo (Turborepo + pnpm) – shipped on AWS EKS via Helm/ArgoCD GitOps with OpenTelemetry tracing.

Tech Stack: TypeScript, Python, Node.js, Mastra, Vercel AI SDK, Claude, OpenAI, Playwright, AWS, Kubernetes, PostgreSQL, MongoDB

PROJECTS

AgentX | *TypeScript, Vercel AI SDK, WebSocket/JSON-RPC*

github.com/adihex/agentx

- Built an event-driven AI agent runtime SDK that models an agentic loop on the Node.js event loop – a 4-phase loop (timers, tool I/O, inference, guards) with non-blocking tools on worker threads so a slow tool call never stalls inference.
- Added ADP, an out-of-band WebSocket/JSON-RPC control plane (Chrome DevTools-style) to pause, inspect, compact memory, and prompt a live agent mid-run, plus a multi-agent orchestrator (dependency graph + plan dispatcher). Sole author; open source.

Zettelkasten Notes | *TypeScript, Turso, React, GCP*

[GitHub](#)

- Built and deployed a note-taking app on top of my AgentX runtime – Turso (libSQL) persistence, authentication, and a React UI, shipped on Google Cloud.

Efficient Keyword Search in CP-ABE | *Java, Cryptography, Searchable Encryption*

B.Tech Thesis

- Designed and implemented (in Java) a searchable-encryption scheme over Ciphertext-Policy Attribute-Based Encryption (CP-ABE) using an inverted-index plus dual-hashmap structure that resolves a query in a single keyword decryption – cutting average search time by 99.83% versus standard ABKS schemes.

TECHNICAL SKILLS

AI & LLM Systems: AI Agents, Multi-Agent Orchestration, Agentic Loops, LLM Evaluation, Prompt Engineering, Foundational Models (Claude, OpenAI, Gemini, Groq), MCP, Vercel AI SDK, Mastra

Languages: TypeScript, JavaScript, Python, Java, SQL, C/C++, Bash

Frameworks & Libraries: Node.js, Express, tRPC, React, OpenTUI, Playwright, Zod, REST APIs

Cloud & Data: AWS (EKS, S3), Docker, Kubernetes, Helm, ArgoCD, Kafka, OpenTelemetry, CI/CD, PostgreSQL, MongoDB, Apache Spark

EDUCATION

Indian Institute of Technology (IIT) Bhubaneswar

B. Tech in Computer Science and Engineering

Nov. 2020 – May 2024

Bhubaneswar, Odisha